

II-Probabilistic Approaches General Tensors, Factored Systems and One-hot encodings

Foundations of Neuro-Symbolic AI

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Outline

- ▶ **Uncertainties**

- ▶ Probability Distributions

Brittleness of Logical Reasoning

Logics cannot handle uncertainties.

But: They are already built in model counts.

Uncertainty: Summarizing the unknown

Model of uncertainties: Not all variables are measureable.

Example: Weather.

Example

Three variables: **Cloud seen**, **Rain today** and **Rain tomorrow**. Let us think of a deterministic relation:

- ▶ If we see a cloud, it will rain today and tomorrow.
- ▶ If we do not see a cloud, it will not rain today and tomorrow.

Now, cloud is not observable, but we want to reason about the probability of rain today and tomorrow.

Logics: Fails, since no entailment of rain today and tomorrow. Probability theory: Works, since we can assign probabilities to the states of rain today and tomorrow, resulting from averages of cloud and not cloud.

Comparing logics and probability theory

Modelling assumptions

- ▶ **Ontological:** What properties does a system have?
→ We assume **fixed sets of categorical properties**.
- ▶ **Epistemological:** What can we tell about these properties?
→ We associate **possibilities or probabilities** with states.

Two main frameworks differing in the **empistemological** assumptions:

- ▶ Logics (**Possibilities**): The traditional mathematical model of intelligence.
- ▶ Probability Theory (**Probabilities**): Handling uncertainties about the systems state.

In both cases, we encode our knowledge in an arrangement of numbers:
Tensors with efficient representations by **Tensor Networks**.

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Recap: Tensors as Propositional Formulas

So far: Tensors with Boolean coordinates and leg dimension 2
Generalize to

- ▶ arbitrary leg dimensions
- ▶ real valued coordinates instead of Booleans $\{0, 1\}$

Probability distributions

Instead of storing 0/1 for possibilities, we can store probabilities (in the interval $[0, 1]$) in the coordinates of a tensor.

Normalization constraint

Probability distributions sum to 1.

We depict this by outgoing legs.